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Human-in-the-loop voice automation system

Abstract

The present disclosure is directed to automating user interactions in an application. In particular, the computing device can determine, based on the input data, an intent associated with performing an action on the computing device. Additionally, the computing device can invoke, using an application-level assistant component configured to provide an application-level interface on the computing device, execution of an application. Moreover, the computing device can transmit, to a system-level assistant component configured to provide a system-level interface with the computing device, instructions to interface with an actionable component of the application. The instructions can be transmitted using a trusted intermediary component configured to facilitate an interface between the system-level assistant component and the application-level assistant component. Furthermore, the computing device can engage, using the system-level assistant component, the actionable component of the application to perform the action.

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Background/Summary

PRIORITY CLAIM

(1) This application is based upon and claims the right of priority under 35 U.S.C. § 371 to International Application No. PCT/US2022/031581, filed on May 31, 2022. International Application No. PCT/US2022/031581 is hereby incorporated by reference herein in its entirety.

FIELD

(2) The present disclosure relates generally to systems and methods for automating tasks on a mobile device using a voice assistant. More particularly, the present disclosure relates to automating user interface actions in a mobile application of a mobile device with human-in-the-loop functionality.

BACKGROUND

(3) Computing devices can automate many tasks and provide various functionality. Users may provide a request to an application-level assistant component using spoken natural language input (e.g., spoken utterances). An application-level assistant component responds to a request by providing responsive user interface output, which can include audible and/or visual user interface output. Application-level assistant components are configured to be interacted with via spoken utterances, such as an invocation indication followed by a spoken query. A client device can include an assistant interface that provides, to a user of the client device, an interface for interfacing with the application-level assistant component (e.g., receives spoken and/or typed input from the user, and provides audible and/or graphical responses), and that interfaces with one or more additional components that implement the application-level assistant component (e.g., remote server device that process user inputs and generate appropriate responses).

SUMMARY

(4) Aspects and advantages of embodiments of the present disclosure will be set forth in part in the following description, or can be learned from the description, or can be learned through practice of the embodiments.

(5) One example aspect of the present disclosure is directed to a computing device. The computing device can include one or more processors, and one or more memory devices storing instructions that are executable by the one or more processors to cause the one or more processors to perform operations. The operations can include determining, based on input data associated with a user of the computing device, an intent associated with performing an action on the computing device. Additionally, the operations can include invoking, using an application-level assistant component configured to provide an application-level interface on the computing device, and execution of an application. Moreover, the operations can further include transmitting, to a system-level assistant component configured to provide a system-level interface with the computing device, instructions to interface with an actionable component of the application. The instructions can be transmitted using a trusted intermediary component configured to facilitate an interface between the system-

level assistant component and the application-level assistant component. Furthermore, the operations can include engaging, using the system-level assistant component, the actionable component of the application to perform the action.

(6) Another example aspect of the present disclosure is directed to a computer-implemented method. The method can include determining, based on input data associated with a user of a computing device, an intent associated with performing an action on the computing device. Additionally, the method can include invoking, using an application-level assistant component configured to provide an application-level interface on the computing device, execution of an application. Moreover, the method can include transmitting, to a system-level assistant component configured to provide a system-level interface with the computing device, instructions to interface with an actionable component of the application. The instructions can be transmitted using a trusted intermediary component configured to facilitate an interface between the system-level assistant component and the application-level assistant component. Furthermore, the method can include engaging, using the system-level assistant component, the actionable component of the application to perform the action.

(7) A further example of the present disclosure is directed to one or more non-transitory computer-readable media. The non-transitory computer-readable media can comprise instructions that when executed by one or more computing devices cause the computing device(s) to perform operations. The operations can include determining, based on input data associated with a user of a computing device, an intent associated with performing an action on the computing device. Additionally, the operations can include invoking, using an application-level assistant component configured to provide an application-level interface on the computing device, and execution of an application. Moreover, the operations can include transmitting, to a system-level assistant component configured to provide a system-level interface with the computing device, instructions to interface with an actionable component of the application. The instructions can be transmitted using a trusted intermediary component configured to facilitate an interface between the system-level assistant component and the application-level assistant component. Furthermore, the operations can include engaging, using the system-level assistant component, the actionable component of the application to perform the action.

(8) In some implementations, the actionable component can be an input field, and the engagement of the actionable component can include inputting a value in the input field. In some implementations, the actionable component can be a rendered output, and the engagement of the actionable component can include parsing rendered output to identify input fields. In some implementations, the actionable component can be an API. In some implementations, the intent can be determined with the application-level assistant component. In some implementations, the system-level assistant component can generate an input comprising data unavailable to the application-level assistant component.

(9) In some implementations, the operations can further include determining missing information and generating a follow-up query. Additionally, the operations can further include the system-level assistant component generating a follow-up query. Furthermore, the operations can further include the system-level assistant component parsing a response to the follow-up query. Moreover, the operations can further include the system-level assistant component initially parsing a response to follow-up query and handing off to the application-level assistant component.

(10) In some implementations, the system-level assistant component can perform an automatic speech recognition (ASR) technique and application-level assistant component can perform a natural-language understanding (NLU) technique.

(11) In some implementations, the operations further include the computing device receiving the content item comprising application-level assistant component. The content item can be obtained from a real-time content selection component of a remote server. For example, a custom chatbot experience for businesses can be performed by the computing device by receiving the content item.

(12) In some implementations, the operations further includes the computing device retrieving, using application-level assistant component, a content item obtained from a real-time content selection component of a remote server, the content item comprising the application.

(13) Another example aspect of the present disclosure is directed to a computing device. The computer device can include a first system, a second system, and a third system. The first system can perform user interface (UI) actions on a plurality of mobile applications. The second system can include a virtual assistant application. The third system can have one or more application program interfaces (APIs) between the first system and the second system. Additionally, the computer device can include one or more processors, and a memory storing instructions that when executed by the one or more processors cause the computer device to perform operations. The device can receive, via an interface of the computing device, a user request having audio data. Additionally, the device can determine, based on the user request, a first query and a second query. Moreover, the device can transmit the first query to a first system. The first query being associated with the first system performing a first UI action on a, first mobile application. Furthermore, the device can transmit, using a third system, a first parameter generated by the first UI action to a second system having a virtual assistant application. The third system includes one or more application program interfaces between the first system and the second system. Subsequently, the device can perform; using the second system, a high-level action based on the second query and the first parameter.

(14) Another example aspect of the present disclosure is directed to a computer-implemented method. The method can include receiving, via an interface of the computing device, a user request having audio data. Additionally, the method can include determining, based on the user request, a first query and a second query. Moreover, the method can include transmitting the first query to a first system. The first query can be associated with the first system performing a first UI action on a first mobile application. Furthermore, the method can include transmitting, using a third system, a first parameter generated by the first UI action to a second system having a virtual assistant application. The third system can include one or more application program interfaces (APIs) between the first system and the second system. Subsequently, the method can include performing, using the second system, a high-level action based on the second query and the first parameter.

(15) A further example of the present disclosure is directed to one or more non-transitory computer-readable media. The non-transitory computer-readable media can comprise instructions that when executed by one or more computing devices cause the computing device(s) to perform operations. The operations include receiving, via an interface of the computing device, a user request having audio data. Additionally, the operations can include determining, based on the user request, a first query and a second query. Moreover, the operations can include transmitting the first query to a first system, the first query being associated with the first system performing a first UI action on a first mobile application. The operations can include transmitting, using a third system, a first parameter generated by the first UI action to a second system having a virtual assistant application. The third system can include one or more application program interfaces (APIs) between the first system and the second system. Subsequently, the operations can include performing, using the second system, a high-level action based on the second query and the first parameter.

(16) In some implementations, the high-level action performed by the second system can be to ask a question. The operations can further include determining, based on the user request, a third query. Additionally, the operations can include receiving, via the interface of the computing device, an answer in response to the question. Moreover, the operations can include modifying the third query based on the answer.

(17) In some implementations, the modification of the third query includes altering one or more parameters being transmitted between the first system and the second system. In some implementations, the modification of the third query includes altering a sequence path associated with subsequent queries to be performed by the first system or the second system.

- (18) In some implementations, the first system has system-level privileges that are higher than privileges associated with the virtual assistant application of the second system. In some implementations, the first system includes a system-level assistant component running as part of an operating system of the computing device. In some implementations, the computing device is a mobile device, and wherein the first system is part of the middleware of the mobile device.
- (19) In some implementations, the first UI action performed by the first system is scrolling and clicking a button on a page of the first mobile application. In some implementations, the high-level action performed by the second system is to open a second mobile application from the plurality of mobile applications. In some implementations, the high-level action performed by the second system is to play a video.
- (20) In some implementations, the operations further include determining, based on the user request, a third query. Additionally, the operations include transmitting, using the third system, a second parameter generated by the second system to the first system. Moreover, the operations include transmitting the third query to the first system, the third query being associated with the first system performing a second UI action on the second mobile application.
- (21) In some implementations, the virtual assistant application is stored in virtual memory associated with a user-space. In some implementations, the parameters transferred between the first system and the second system are only transmitted using the third system.
- (22) In some implementations, the trusted intermediary component includes a middleware API. In some implementations, the trusted intermediary component includes an operating system kernel API. In some implementations, the trusted intermediary component includes a direct remote procedure call (RPC)-based API.
- (23) In some implementations, the computing device can obtain user consent before the first system performs the first UI action on the first mobile application. Additionally, the computing device can display, while the first system is performing the first UI action, an indicator on a display of the computing device that the first system is controlling the computing device.
- (24) Another example of the present disclosure is directed to a computing device that includes one or more processors, and one or more memory devices storing instructions that are executable by the one or more processors to cause the one or more processors to perform operations. The computing device can determine, based on input data associated with a user of the computing device, an intent associated with performing an action on the computing device. The computing device can be invoked, using an application-level assistant component (e.g., application-level assistant component **220**) configured to provide an application-level interface on the computing device, execution of an application (e.g., mobile application). The computing device can transmit, to a system-level assistant component (e.g., automation service) configured to provide a system-level interface with the computing device, instructions to interface with an actionable component of the application (e.g., mobile application). The instructions can be transmitted using a trusted intermediary component configured to facilitate an interface between the system-level assistant component and the application-level assistant component. The computing device can engage, using the system-level assistant component (e.g., automation system), the actionable component of the application to perform the action.
- (25) Other aspects of the present disclosure are directed to various systems, apparatuses, non-transitory computer-readable media, user interfaces, and electronic devices.
- (26) These and other features, aspects, and advantages of various embodiments of the present disclosure will become better understood with reference to the following description and appended claims. The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate example embodiments of the present disclosure and, together with the description, serve to explain the related principles.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Detailed discussion of embodiments directed to one of ordinary skill in the art is set forth in the specification, which makes reference to the appended figures, in which:
- (2) FIG. 1 depicts a block diagram of an example computing system that performs automation techniques according to example embodiments of the present disclosure;
- (3) FIG. 2 depicts an example automation system according to example embodiments of the present disclosure;
- (4) FIG. 3 depicts example environment in which an automation system is invoked by a client device or a user according to example embodiments of the present disclosure;
- (5) FIG. 4 depicts a user experience (UX) of a query example for adding an item to a shopping application according to example embodiments of the present disclosure;
- (6) FIG. 5 depicts a flow chart diagram of an example method to automating an action to be performed on a mobile application according to example embodiments of the present disclosure;
- (7) FIG. 6 depicts a flow chart diagram of an example method of a human-in-the-loop implementation for inputting a missing parameter, according to example embodiments of the present disclosure;
- (8) FIG. 7 depicts a flow chart diagram of another example method to automating an action to be performed on a mobile application according to example embodiments of the present disclosure;
- (9) FIG. 8 depicts a flow chart diagram of an example method of a human-in-the-loop implementation for modifying a query, according to example embodiments of the present disclosure;
- (10) FIG. 9 depicts a flow chart diagram of an example method to securely transfer data between two mobile applications, according to example embodiments of the present disclosure.

DETAILED DESCRIPTION

- (11) A user can interact with an application-level assistant component (e.g., virtual assistant, automated assistant, interactive personal assistants, intelligent personal assistants) by providing user input to the automation system after the automation system has been invoked. In some instances, the user interface inputs that can invoke the automation system via a client device include a hardware and/or virtual button at the client device for invoking the automation system (e.g., a tap of a hardware button, a selection of a graphical interface element displayed by the client device). The automation system can additionally or alternatively be invoked in response to one or more spoken invocation phrases, such as trigger words.
- (12) The user input can be a natural language input (e.g., spoken utterance) associated with a request. For instance, automatic speech recognition and/or natural language understanding of the spoken utterance can be performed by the automation system. Automatic speech recognition (ASR) can be performed by the automation system in order to detect certain terms that can correspond to particular actions capable of being performed. Alternatively, or additionally, natural language understanding (NLU) can be performed by the automation system in order to identify certain intent(s) capable of being performed. The automation system can process the request and respond to the request with a user interface output. Additionally, a user may want to interact with the automation system to operate other applications (e.g., third-party applications installed on a client device).
- (13) As application-level assistant component (e.g., virtual assistant) becomes an important part of mobile user experience, techniques described herein can increase the capabilities of an application-level assistant component (e.g., what users can do using an application-level assistant component on smartphones) to improve the mobile user experience. The automation system enables fulfillment of actions performed on a mobile application, while also reducing the feature development and

maintenance cost by reusing the existing functionalities in a large number of mobile applications.

(14) In some implementations, the automation system enables voice-centric mobile application automation to significantly extend the fulfillment capabilities of a virtual assistant with mobile applications. voice-centric mobile application automation includes certain tasks that are automatable (e.g., clicking and filling in forms, sending data to another user). The automation tasks performed by the automation system include a series of user interface (UI) actions. Such automation tasks can be classified into three subtypes: (1) tasks that the user wants to perform manually; (2) tasks that the user wants to supervise; and (3) tasks that the user does not want to supervise. The automation system can automatically perform tasks that are associated with subtypes (2) and (3).

(15) Both subtypes (2) and (3) can require human-in-the-loop-style execution (e.g., implementation). For example, subtype (2) may require human-in-the-loop implementation because the user may want to stop and alter the process. Subtype (3) may require human-in-the-loop implementations when as part of execution the automation flow, the automation system determines that one or more parameter values are needed to be obtained from the user. The automation system enables human-in-the-loop voice automation for mobile applications. Unlike a conventional end-to-end automation, human-in-the-loop automation enables users to alter the process flows and interact with content of the mobile application, while a vast majority of the tedious tasks can be automated by the automation system.

(16) Example embodiments according to aspects of the present disclosure relate to automating user interface actions in a mobile application of a mobile device using an automation system. The automation system can include an application-level assistant component, a system-level assistant component, and a trusted intermediary component to interface between the application-level assistant component and the system-level assistant component. In one example, a user can request the automation system to obtain data from a first mobile application (e.g., photo application) and transmit the obtained data to a person named 'John' using a second mobile application (e.g., email application). The automation system can automate the process by executing scripts on each of the mobile applications and present the user interface actions to the user as each script is being executed. This allows the user the ability to change or modify the process in real-time while viewing the user interface actions. Continuing with the example, while viewing the script being executed to automatically send the data using the second mobile application, the user may decide that the automation process is sending the data to a different 'John,' so may modify the process by selecting the correct 'John' to send the data. Alternatively, or additionally, the automation system can ask the user a clarifying question regarding which 'John' to send the data, when the automation system determines that there are multiple people named 'John.'

(17) The automation system described herein differs from conventional website automation systems, because conventional website automation systems specifically targets web browsers. In contrast, the automation system **200** enables the automation of a query fulfillment process on mobile applications of a mobile device. Conventional website automation systems are designed to interface with web pages (e.g., HTMLs), while automation system **200** is designed to interface with mobile applications and execute actions on the user interface of a mobile application.

(18) Additionally, the automation system described herein is different from conventional systems that automate recurring activities or queries. Automating a recurring mobile application activity or query can include suggesting an automated action based on a similar context (e.g., day of week, time of day, and location). However, both recurring queries and activities are for one action and does not automate user journeys across a plurality of mobile applications or a plurality of actions in one mobile application.

(19) The technology described herein can provide a number of technical effects and benefits. For example, as previously indicated, the technology described herein can improve security, partly because the execution of the UI scripts are handled by the system-level assistant component, which

can be a system application that is running in a private compute core (PCC) sandbox. Additionally, the application-level assistant component can interact with the system-level assistant component by encapsulating the context and UI actions of the mobile application as augmented built-in intents (BIIs). The augmented BIIs can be coarse-grained descriptions of actions and thus are allowed, security-wise, by the platform of the mobile device. Additionally, by having the human-in-the-loop interface, the automation system provides additional safety checks such as the user consent dialog, UI indicator during the script execution, and system settings. Furthermore, the automation system (e.g., by using the trusted intermediary component) may track the interactions between system-level assistant component and the application-level assistant component such that it automatically triggers the system security features in real-time (e.g., starting to show the indicator when the script execution is started) if an issue is detected.

(20) With reference now to the figures, example embodiments of the present disclosure will be discussed in further detail.

(21) Example Devices and Systems

(22) FIG. 1 depicts a block diagram of an example computing system **100** that performs automation techniques using an automation system according to example embodiments of the present disclosure. The system **100** can include a user computing device **102** and a server computing system **130** that can be communicatively coupled over a network **180**.

(23) The user computing device **102** can be any type of computing device, such as, for example, a personal computing device (e.g., laptop or desktop), a mobile computing device (e.g., smartphone or tablet), a gaming console or controller, a wearable computing device, an embedded computing device, a smart device (e.g., smart television, smart appliance), a virtual reality system, an augmented reality system, or any other type of computing device.

(24) The user computing device **102** can include one or more processors **112** and a memory **114**. The one or more processors **112** can be any suitable processing device (e.g., a processor core, a microprocessor, an ASIC, an FPGA, a controller, a microcontroller, etc.) and can be one processor or a plurality of processors that are operatively connected. The memory **114** can include one or more non-transitory computer-readable storage media, such as RAM, ROM, EEPROM, EPROM, flash memory devices, magnetic disks, etc., and combinations thereof. The memory **114** can store data **116** and instructions **118** which are executed by the processor **112** to cause the user computing device **102** to perform operations.

(25) In some implementations, the user computing device **102** can store or include an automation system **120**. An example automation system **120** is further discussed in FIG. 2.

(26) Additionally, or alternatively, an automation system **140** can be included in or otherwise stored and implemented by the server computing system **130** that communicates with the user computing device **102** according to a client-server relationship. For example, the automation system **120** can be implemented by the server computing system **140**. Thus, the automation system **120** can be stored and implemented at the user computing device **102** and/or the automation system **140** can be stored and implemented at the server computing system **130**. In some implementations, the outputs from the automation system **140** can be received from the server computing system **130** over network **180**, stored in the user computing device memory **114**, and then used or otherwise implemented by the one or more processors **112**.

(27) The user computing device **102** can also include one or more user input components **122** (e.g., keyboard, microphone, camera) that receives user input. For example, the user input component **122** can be a touch-sensitive component (e.g., a touch-sensitive display screen or a touch pad) that is sensitive to the touch of a user input object (e.g., a finger or a stylus). Other example user input components include a microphone, a camera, or other means by which a user can provide user input (e.g., audio data, video data).

(28) Some user interface inputs that can invoke the automation system **120** via the user computing device **102** include a hardware and/or virtual button at the user computing device **102** for invoking

the automation system **120** (e.g., a tap of a hardware button, a selection of a graphical interface element displayed by the client device). Many automation system **120** can additionally or alternatively be invoked in response to one or more spoken invocation phrases, such as trigger words. Also, for example, the automation system **120** may be invoked based on one or more gestures of the user, such as pressing a button on a device and/or motioning in a particular manner such that the motion can be captured by a camera of a device.

(29) The server computing system **130** can include one or more processors **132** and a memory **134**. The one or more processors **132** can be any suitable processing device (e.g., a processor core, a microprocessor, an ASIC, an FPGA, a controller, a microcontroller, etc.) and can be one processor or a plurality of processors that are operatively connected. The memory **134** can include one or more non-transitory computer-readable storage media, such as RAM, ROM, EEPROM, EPROM, flash memory devices, magnetic disks, etc., and combinations thereof. The memory **134** can store data **136** and instructions **138** which are executed by the processor **132** to cause the server computing system **130** to perform operations.

(30) In some implementations, the server computing system **130** can include or be otherwise implemented by one or more server computing devices. In instances in which the server computing system **130** includes plural server computing devices, such server computing devices can operate according to sequential computing architectures, parallel computing architectures, or some combination thereof. As described above, the server computing system **130** can store or otherwise include an automation system **140**. An example automation system **140** is further described in FIG. 2.

(31) The user computing device **102** and/or the server computing system **130** can train the automation system **120** and/or **140** via interaction with the training computing system **150** that can be communicatively coupled over the network **180**. The training computing system **150** can be separate from the server computing system **130** or can be a portion of the server computing system **130**.

(32) The training computing system **150** includes one or more processors **152** and a memory **154**. The one or more processors **152** can be any suitable processing device (e.g., a processor core, a microprocessor, an ASIC, an FPGA, a controller, a microcontroller, etc.) and can be one processor or a plurality of processors that are operatively connected. The memory **154** can include one or more non-transitory computer-readable storage media, such as RAM, ROM, EEPROM, EPROM, flash memory devices, magnetic disks, etc., and combinations thereof. The memory **154** can store data **156** and instructions **158** which are executed by the processor **152** to cause the training computing system **150** to perform operations. In some implementations, the training computing system **150** includes or is otherwise implemented by one or more server computing devices.

(33) The training computing system **150** can include a model trainer **160** that trains the automation system **120** and/or **140** stored at the user computing device **102** and/or the server computing system **130** using various training or learning techniques. In particular, the model trainer **160** can train the automation system **120** and/or **140** based on a set of training data **162**. The training data **162** can include, for example, previous user requests, answers in response to the user requests, clarifying questions, and actions automatically performed on the mobile applications.

(34) In some implementations, if the user has provided consent, the training examples can be provided by the user computing device **102**. Thus, in such implementations, the automation system **120** provided to the user computing device **102** can be trained by the training computing system **150** on user-specific data received from the user computing device **102**. In some instances, this process can be referred to as personalizing the automation system.

(35) The model trainer **160** can include computer logic utilized to provide desired functionality. The model trainer **160** can be implemented in hardware, firmware, and/or software controlling a general-purpose processor. For example, in some implementations, the model trainer **160** can include program files stored on a storage device, loaded into a memory and executed by one or

more processors. In other implementations, the model trainer **160** can include one or more sets of computer-executable instructions that are stored in a tangible computer-readable storage medium such as RAM, hard disk, or optical or magnetic media.

(36) The network **180** can be any type of communications network, such as a local area network (e.g., intranet), wide area network (e.g., Internet), or some combination thereof and can include any number of wired or wireless links. In general, communication over the network **180** can be carried via any type of wired and/or wireless connection, using a wide variety of communication protocols (e.g., TCP/IP, HTTP, SMTP, FTP), encodings or formats (e.g., HTML, XML), and/or protection schemes (e.g., VPN, secure HTTP, SSL).

(37) The automation system **120** and/or **140** described in this specification may be used in a variety of tasks, applications, and/or use cases. In some implementations, the input to the automation system **120** and/or **140** of the present disclosure can be speech data, text, or natural language data. The automation system **120** and/or **140** can process the speech data, text, or natural language data to generate an output.

(38) FIG. 1 illustrates one example computing system that can be used to implement the present disclosure. Other computing systems can be used as well. For example, in some implementations, the user computing device **102** can include the model trainer **160** and the training dataset **162**. In such implementations, the automation system **120** can be both trained and used locally at the user computing device **102**. In some of such implementations, the user computing device **102** can implement the model trainer **160** to personalize the automation system **120** based on user-specific data.

(39) FIG. 2 depicts an example automation system according to example embodiments of the present disclosure. The automation system **200** can be an example of the automation system **120** and/or automation system **140** in FIG. 1. The automation system **200** can include a system-level assistant component **210**, an application-level assistant component **220**, and a trusted intermediary component **230**. The trusted intermediary component **230** can be one or more application program interfaces (APIs) between the system-level assistant component **210** and the application-level assistant component **220**.

(40) The system-level assistant component **210** can run as part of the system software (e.g., operating system or middleware). The system-level assistant component **210** can have extra privileges that are not typically available in user-space mobile applications. The application-level assistant component **220** can be a user-space virtual assistant application. The trusted intermediary component **230** can include middleware API, operating system kernel API, and/or direct RPC (remote procedure call)-based mechanisms.

(41) In terms of execution (i.e., performer), the system-level assistant component **210** can perform low-level UI actions (e.g., scrolling and clicking a button). The application-level assistant component **220** can perform or be requested to perform high-level UI actions (e.g., opening an app or playing a video).

(42) In terms of natural language process (NLP), the system-level assistant component **210** can include an automatic speech recognition (ASR), natural language understanding (NLU), and fulfillment logic for the low-level UI actions (e.g., low-level commands). In some instances, the system-level assistant component **210** may not have an ASR. The application-level assistant component **220** can include an ASR, NLU, and fulfillment logic by default for the high-level UI actions (e.g., high-level commands).

(43) With regards to handover options, the automation system **200** can include a plurality of handover embodiments.

(44) In a first handover embodiment, a user can use the application-level assistant component **220** (e.g., virtual assistant) to start a session (e.g., long-press home, hot-word triggering) and send an explicit query (e.g., start a sandbox mode) to handover to the NLU of the system-level assistant component **210**.

(45) In a second handover embodiment, a user can use the system-level assistant component **210** to start a session (e.g., a button on the system UI or an integrated application). If the query is not understood or fulfilled by the NLU of the system-level assistant component **210**, then the system-level assistant component **220** can transmit the query to the NLU of the application-level assistant component **220** for processing. Continuing with this second embodiment, the system-level assistant component **210** uses the trusted intermediary component **230** to transmit the dictated query to the NLU of the application-level assistant component **220** in order to bypass the ASR of the application-level assistant component. This can improve the user experience by enabling the dictated query to be transmitted to the application-level assistant component without having to ask the user again for the request.

(46) In a third handover embodiment, a user can use the system-level assistant component **210** to start a session and send an explicit query to hand over to the NLU of the application-level assistant component **220**.

(47) In a fourth handover embodiment, a user can use the application-level assistant component **220** to start a session. If the query is not understood or fulfilled by the NLU of the application-level assistant component **220**, then the application-level assistant component **220** can transmit the query to the system-level assistant component **210**. The NLU or fulfillers of system-level assistant component **210** can execute (e.g., perform) the request for certain types of queries. In this fourth handover embodiment, the application-level assistant component **220** can use the trusted intermediary component **230** to send the dictated query to NLU of the system-level assistant component **210** in order to bypass the ASR of the system-level assistant component **210**. In some instances, the application-level assistant component **220** can use the trusted intermediary component **230** to send the dictated query to fulfillers of the system-level assistant component **210** in order to bypass both the ASR and NLU of the system-level assistant component **210**.

(48) With regards to types of journeys, the system-level assistant component can perform both multi-application journeys and multi-turn journeys in the same mobile application.

(49) In the multi-application journey example, an automation can consist of a sequence of tasks for multiple mobile applications. The automation system **200** can automate the journey using a hierarchical approach. For example, the application-level assistant component **220** (e.g., user-space virtual assistant) can handle the inter-application journeys (e.g., opening a mobile application, sending an intent, or passing parameters between mobile applications), while the system-level assistant component **210** can handle the intra-application journeys (e.g., performing one or more UI actions on a specific activity of a mobile application, performing one or more UI actions on a specific page of a mobile application). By bifurcating the inter-application journeys and the intra-application journeys, the automation system can greatly improve the scalability and maintainability of the automation techniques described herein. For example, let assume that three mobile applications are used as part of an automation task, and for each mobile application there are five different versions which may require slightly altered scripts. Using the techniques described herein, this example may need a maximum of 15 total scripts (i.e., 3 different mobile applications $\times$ 5 different versions for each of the mobile applications=15). In contrast, conventional automation techniques may require 125 total scripts (i.e., $5^{\text{circumflex over ()}}3=5*5*5=125$) because one script may be required to be written per automation task. Additionally, the 15 scripts that are developed for the hierarchical automation of the automation system **200** are reusable for other automation tasks that involve the same mobile applications, while the 125 scripts that are written for the conventional automation techniques are not reusable.

(50) In the multi-turn journey example, the automation system **200** is able to obtain user inputs at runtime and process the user inputs at runtime. The core of human-in-the-loop automation can be that the automation system **200** can get the user inputs (e.g., via system UI) and process the user inputs at runtime. For example, the automation system **200** can use the user input data to derive the parameters for the following (e.g., subsequent) tasks. The automation system **200** can also send the

data back to the application-level assistant component **220** (e.g., user-space virtual assistant application) so that the application-level assistant component **220** can alter the following tasks or ask relevant clarification questions to the user. In some implementations, private data (due to the privacy requirement) may only be available in the system-level assistant component **210** and thus cannot be transmitted to the application-level assistant component **220** (e.g., have to be kept in the system-level assistant component **210**). In these implementations, the availability of such private data in the system-level assistant component **210** is known to the application-level assistant component **220**, and as a result the application-level assistant component **220** can skip the derivation process to determine or obtain the private data from the user (e.g., not asking any clarification questions). Instead, the fulfillment command of the application-level assistant component **220** can specify to use the private data available in the system-level assistant component **210**.

(51) In some implementations, the system-level assistant component **210** (also referred as a first system) enables an application-level assistant component **220** (also referred as a second system) to perform any tasks (e.g., low-level UI actions) on mobile application **240** of a mobile device that the user can manually perform. The automation system **200** can utilize the system-level assistant component **210** that is running as a system application in order to execute user interface (UI) scripts **270** based on received user input **205**. Additionally, the automation system **200**, by using the trusted intermediary component **230**, can utilize the system framework that bridges between application-level assistant component **220**, the system-level assistant component **210**, and the target third-party applications (e.g., mobile application **240**).

(52) In some implementations, upon an invocation (e.g., based on user input **205**) of the application-level assistant component **220**, the system-level assistant component **210** uses the trusted intermediary component **230** (e.g., voice interaction session (VIS) API) to retrieve context information from a page (e.g., first page **245**, second page **250**, and so on) of a mobile application **240** (e.g., third-party mobile application installed on a mobile device). For example, the first page **245** can be the home page of the mobile application **240**, and the second page **250** can be the payment page of the mobile application. In some instances, the context information is retrieved from an activity (e.g., first activity **255**, second activity **260**, and so on) of the mobile application **240**. For example, when the mobile application **240** is an email application, the first activity **255** can be sending of an email to a recipient, and the second activity **260** can be reading aloud a received email from a sender.

(53) In some implementations, the capability and entity information of the mobile application **240** can be obtained by the system-level assistant component **210** running as a system application on a private compute core (PCC). The system-level assistant component **210** can send the context information obtained from the mobile application to an on-screen understanding engine. The on-screen understanding engine can derive context information to generate UI scripts **270**. The UI scripts **270** enables the automation system **200** to perform an automated action associated with a page or activity of the mobile application **240**. Additionally, the derived context information can be encoded in the form of a direct action, which can be then sent to the speech recognizer, and/or fulfillment pipelines of the application-level assistant component **220**.

(54) Subsequently, the fulfilled results can be sent to performers to open the page of the mobile application **240** activity and execute a script from the UI scripts **270**. The script execution relies on the system-level assistant component **210**, which can be invoked by using the trusted intermediary component **230** (e.g., VIS API). The system-level assistant component **210** can obtain the respective script from the online infrastructure storing the UI scripts **270** and execute the script using a trusted intermediary component **230** (e.g., system API) that can drive certain UI actions on the mobile application **240**.

(55) The automation system **200** enables the automated fulfillment of actions in various mobile applications which have not been integrated with the mobile platform's application actions APIs.

For example, the automation system **200** enables the fulfillment of actions without the need for the third-party developers of the mobile application to use the mobile platform's software development kit (SDK) to enable other applications on the mobile platform to execute an action on the mobile application.

(56) In some implementations, the automation system **200** can automate UI actions associated with activities (e.g., capabilities) of a mobile application. The automation system **200** can use an application integration platform for built-in intents (BIIs) and NLU. The automation system **200** can include a set of fulfillers, a performer, and other associated platform modules of the application-level assistant component **220** to process user queries. For example, the automation system **200** is able to process user queries, even without a mobile application having been integrated with the platform's application actions APIs (e.g., actions are explicitly declared and supported by the mobile application), because the techniques described herein can extract the third-party application capabilities based on screen understanding of the different pages in the mobile application to enable UI script-based execution for third-party mobile applications. These techniques allow the automation system **200** to enable increasing the application-level assistant component **220** fulfillment capabilities. Furthermore, the system-level assistant component **210** can increase fulfillment capabilities in a mobile application without requiring any direct code changes in the mobile application. Given that the system-level assistant component **210** can have system-level privileges, the system-level assistant component **210** can rely on the system platform of the computing device to execute UI scripts in a mobile application.

(57) Additionally, the automation system **200** can use augmented built-in intents (BIIs) **280** that have been extracted from the mobile application **240**. In conventional application-level assistant component systems, an application-level assistant component can only perform certain actions on the mobile application that are explicitly declared and supported by the mobile application. By using augmented BIIs **280**, the automation system **200** can extract from the mobile application **240** (e.g., by using the screen contexts) certain actions that can be performed, without the need for the mobile application to explicitly declare such actions. In some implementations, the system-level assistant component **210**, which can be run as part of the mobile platform, can obtain (e.g., retrieve, receive) the augmented BIIs **280** for processing. The system-level assistant component **210** can include a dedicated system-level assistant component performer and modularized fulfillers for system-level assistant component **210**. During implementation, the fulfiller of the system-level assistant component and other application related fulfillers can be tightly integrated depending on the data source and execution abstraction. The advantages of such dedicated fulfillers and performers in the system-level assistant component **210** include improved modularity, fault isolation, testability, and so on.

(58) Moreover, the automation system **200** can process the user input **205** to generate one-time queries, explicit queries, follow-on queries, conversation-style queries, and other types of queries. Examples of one-time queries are “show sports news on <a news app>” and “send a message to <contact> on <a 3P messenger>”.

(59) The automation system **200** can fulfill queries by using a trusted intermediary component **230**. The trusted intermediary component **230** can include a platform API to retrieve the augmented BIIs **280** associated with an activity (e.g., add an item to a shopping cart, purchase item(s) in a shopping cart) of a mobile application **240**. Additionally, the platform API can execute the selected augmented BII **280** on the mobile application **240**. For example, the platform API can rely on the system-level assistant component **210** that is run on a Private Compute Core (PCC). Given that the system-level assistant component **210** can run in a secure sandbox, the system-level assistant component **210** can have the requisite permissions to access the screen context and run the screen understanding modules so as to derive a list of augmented BIIs **280** from the screen context of the mobile application. The system-level assistant component **210** can, optionally, extract parameter values (e.g., entities or categorized texts on the screen) and return them as part of the BII values to

the application-level assistant component **220**.

(60) The system-level assistant component **210** is able to retrieve the augmented BII **280** in such a way that it does not affect the user experience (e.g., in terms of UI rendering and query processing latencies). The system-level assistant component **210** can also capture the augmented BII data (i.e., serializable) as part of user context and flow it through the platform of the application-level assistant component **220** (e.g., for ASR, NLU, and fulfillment), without affecting any existing on-device or server-side NLP flows. The automated system **200** also minimizes code duplication and allows to build highly maintainable infrastructure that can be maintained by the existing virtual assistant infrastructure in an efficient way.

(61) In one implementation, the user query can specify a target mobile application that is not in the foreground of the display of the mobile device by using an explicit query. In this implementation, the system-level assistant component **210** uses a backend pipeline that mines the augmented BII **280** and passes the augmented BII **280** to the on-device platform of application-level assistant component **220**. The system-level assistant component **210** can transmit the augmented BII **280** into the on-device stack (e.g., POP) of the application-level assistant component **220** so that the augmented BII information can be used as part of the processing.

(62) In some implementations, the trusted intermediary component **230** (e.g., DirectAction API) can call the system-level assistant component **210** if the user input **205** is associated with an augmented direct action. Upon invocation, the system-level assistant component **210** fetches a relevant UI script from the UI scripts **270** and executes the relevant UI script on an activity and/or page of a mobile application.

(63) As part of the human-in-the-loop implementations, the system-level assistant component **210** can obtain user consent before executing the relevant UI script. Additionally, the system-level assistant component **210** can show a UI indicator on the screen while executing the relevant UI script. Subsequently, one or more parameters from the execution result can be returned to the application-level assistant component **220** via the trusted intermediary component **230**.

(64) The application-level assistant component **220** can have a platform that is able to fulfill queries from the system-level assistant component using information from the augmented BII **280**. For example, the application-level assistant component **220** can generate a ClientOp that can be used to direct a performer, which then can call a trusted intermediary component (e.g., direct actions API) with an augmented direct action argument value.

(65) Additionally, the application-level assistant component **220** can obtain the augmented BII **280** by using the trusted intermediary component **230** (e.g., direct actions API, context API). The information from the augmented BII **280** associated with an activity or page of the mobile application can be obtained in a way that does not delay any of the subsequent operations.

(66) The application-level assistant component **220** can include fulfillers to perform explicit queries (e.g., ones that require opening an app or sending an intent with some specific parameter values before executing a script) by issuing multiple ClientOps. The application-level assistant component **220** can perform multi-app journeys (e.g., open app 1, replay script A, open app 2, and replay script B) in the near future. The fulfiller framework of the application-level assistant component **220** can rank the generated ClientOps or SemOps using provided product requirements. The product requirement can be updated as the relevant products introduce more use cases. Additionally, the ranking can be updated and/or modified by the training computing system **150** in order to optimize the fulfillment process.

(67) Partly due to enhanced security measures, the trusted intermediary component **230** enables data (e.g., context and user data) to be transmitted from the system-level assistant component **210** to the application-level assistant component **220**, and data to be transmitted from the application-level assistant component **220** to the system-level assistant component **210**. Therefore, in some implementations, the data cannot be transmitted directly from the system-level assistant component **210** to the application-level assistant component **220** or from the application-level assistant

component **220** to the system-level assistant component **210**. For example, data (e.g., contact names) that are obtained from a mobile application **240** by the system-level assistant component **210** can be transmitted, via the trusted intermediary component **230**, to the application-level assistant component **220** as part of the augmented BILs **280**.

(68) Moreover, in order to improve performance and user experience, the application-level assistant component **220** and the system-level assistant component **210** can work together to ensure that the user consent is checked at the right moment, the relevant user opt-in flags are stored and used, and proper UI indicators are shown during any script executions. Some of such logic can be built in the automation system **200**, while the triggering and control of such logic can be performed by the application-level assistant component **220**.

(69) Continuing with the human-in-the-loop implementations, the automation system **200** can check whether any of the required parameter values are missing in the given query and context information, and as a result ask clarification queries (e.g., “what do you want to search for?” for the user query “search on a social media application”) and getting the missing information ahead of time even before the query can be fulfilled. As previously discussed, some of the contextual data may be sensitive and thus will be kept in the system-level assistant component **210** and not be shared with the application-level assistant component **220**. The NLU can be expected to understand such data types and rely on the system-level assistant component **210** for filling in the parameter value fields.

(70) Additionally, the application-level assistant component **220**, using techniques described herein, can handle follow-on voice queries that are issued by the user without starting a new conversation. Without having to make the user re-invoke the application-level assistant component **220** for follow-on queries, the automation system can improve the user experience. This is beneficial because many of the user journeys involve conversation-style dialogues (e.g., taking notes or sharing photos).

(71) Referring to FIG. 3, an example environment **300** is provided which includes an automation system **310**, **345** (e.g., automation system **120**, automation system **140**, automation system **200**) that may be invoked by a user **360** or a client device **340** (e.g., user computing device **102** in FIG. 1). In one implementation, the application-level assistant component **220** in FIG. 2 can be an example of the automation system **310**. The environment includes a standalone interactive speaker **305** with a microphone **315** and a camera **320**. The speaker **305** may be executing, at least in part, an automation system **310** that may be invoked with an invocation phrase. In the example environment, the user **360** can speak a spoken utterance (e.g., “Okay Assistant, send a photo to person A”) in proximity to the speaker **305**. If the automation system **310** is configured to be invoked by the phrase “Okay Assistant,” the invoked assistant may process the query that follows the invocation phrase (i.e., “send a photo to person A”). In some implementations, the automation system **310** can be capable of being invoked by the user performing one or more actions that can be captured by the cameras of the automation system **310**. For example, automation system **310** can be invoked by the user looking in the direction of automation system **310**, making a waving motion in the direction of automation system **310**, and/or one or more other actions that can be captured by the camera of automation system **310**.

(72) Additionally, a client device **340** (e.g., user computing device **102** in FIG. 1, mobile device) can include an automation system **345** (e.g., automation system **120**, automation system **140**, automation system **200**) that may be invoked by a user **360** or the speaker **305**. The client device **340** can execute the automation system **345**. The automation system **345** can be invoked by uttering an invocation phrase proximate to the client device **340** such that the audio may be captured by a microphone **350** and/or performing an action that may be captured by camera **355**. Moreover, the automation system **345** can be invoked based on a user input (e.g., user pressing a button **365**, user pressing an icon displayed on a graphical user interface **360**).

(73) In some implementations, user **360** may invoke the automation system **345** by pressing an icon

displayed on the graphical user interface **360** to invoke the automation system **345**. For example, pressing a first icon can invoke the application-level assistant component (e.g., application-level assistant component **220** of the automation system **200**), and pressing a second icon can invoke the system-level assistant component (e.g., system-level assistant component **210** in automation system **200**).

(74) In some implementations, user **360** may invoke the automation system **310** and/or automation system **345** by uttering a first keyword associated with the application-level assistant component (e.g., application-level assistant component **220** of the automation system **200**), or uttering a second keyword associated with system-level assistant component (e.g., system-level assistant component **210** in automation system **200**). Further, user **360** may invoke the application-level assistant component by performing a first action and invoke the system-level assistant component by performing a second action. Based on which invocation phrase is uttered and/or which action is performed, the user **360** can indicate which of the multiple assistants that are executing on speaker **305** and/or client device **340** that the user **360** has interest in processing a spoken query. In some implementations, one or more of the automation systems of FIG. 3 may be absent. Further, the example environment may include additional automation systems that are not present in FIG. 3.

(75) Each of the automation systems **310**, **345** can include one or more components of the automation system described in FIGS. 1-2. For example, automation systems **310**, **345** may include its own speech capture component to process incoming queries, visual capture component to process incoming visual data, hot word detection engine, and/or other components. In some implementations, the system-level assistant component **210** and the application-level assistant component **220** that are executing on the same device can share one or more components that may be utilized by both of the components. For example, the system-level assistant component **210** and the application-level assistant component **220** may share an on-device speech recognizer, on-device NLU engine, and/or one or more of the other components.

(76) The automation system **310**, **345** may be configured to process invocation input, such as an utterance that includes the phrase “OK Assistant” or other invocation input, which may indicate that the user has interest in providing a query. The automation system **310**, **345** can include multiple components for processing a query, once invoked, for example, a local speech-to-text (“STT”) engine (that converts captured audio to text), a local text-to-speech (“TTS”) engine (that converts text to speech), a local natural language processor (that determines semantic meaning of audio and/or text converted from audio), and/or other local components. Because the client devices executing application-level assistant components may be relatively constrained in terms of computing resources (e.g., processor cycles, memory, battery, etc.), the local components may have limited functionality relative to any counterparts that are included in any cloud-based application-level assistant component components that are executing remotely in conjunction with the application-level assistant component(s).

(77) In some implementations, the automation system **310**, **345** may be invoked by one or more gestures that indicate that the user has interest in interacting with the primary application-level assistant component. For example, a user may demonstrate intention to invoke an application-level assistant component by interacting with a device, such as pressing a button or a touchscreen, perform a movement that is visible and may be captured by an image capture device, such as camera, and/or may look at a device such that the image capture device can recognize the user movement and/or positioning. When a user performs a gesture or action, the application-level assistant component may be invoked and begin capturing audio data that follows the gesture or action, as described above. Further, as described above, multiple application-level assistant components may be invoked by the same invocation input such that a particular gesture may be a common invocation to more than one application-level assistant component.

(78) Example Graphical User Interfaces

(79) FIG. 4 depicts a user experience (UX) of a query example for adding an item to a shopping

application according to example embodiments of the present disclosure. As previously mentioned, the developers of the third-party application have not integrated (e.g., by implementing Direct Action handlers) the third-party application (e.g., shopping application) with the mobile platform so that other applications (e.g., application-level assistant component) can perform an action on the mobile application. Therefore, using conventional techniques, the application-level assistant component cannot fulfill the user's request to add an item in the shopping application. By using the techniques described in FIG. 2, the automation system **200** can receive a user request at **410** and initiate a voice control mode at **420**. In response to receiving the user request at **410**, the automation system (e.g., automation system **120**, automation system **140**, automation system **200**, automation system **310**, automation system **345**) can obtain a UI script (e.g., from the plurality of UI scripts **270** in FIG. 2) to execute the action at **430**. and fulfill the request at **430**. During operation **430**, the automation system can display to the user that the automation system is controlling the computing device of the user (i.e., as depicted at **435**). Once the action at **430** is executed by the automation system, the automation system can give control back to the user (as part of the human-in-the-loop implementations) at **440** in order for the user to confirm (e.g., complete) the action (e.g., purchase the items in the shopping cart).

Example Methods

(80) FIG. 5 depicts a flow chart diagram of an example method for automating an action to be performed on an application according to example embodiments of the present disclosure. Although FIG. 5 depicts steps performed in a particular order for purposes of illustration and discussion, the methods of the present disclosure are not limited to the particularly illustrated order or arrangement. The various steps of method **500** can be omitted, rearranged, combined, and/or adapted in various ways without deviating from the scope of the present disclosure.

(81) In some implementations, a computing device (e.g., user computing device **102**, server computing system **130**, automation system **200**, speaker **305**, client device **340**) can include an automation system (e.g., automation system **200**). The automation system **200** can include a system-level assistant component **210**, an application-level assistant component **220**, and a trusted intermediary component **230**. Additionally, the computing device can include one or more processors, and a memory storing instructions that when executed by the one or more processors cause the computer device to perform operations described in method **500**.

(82) At **502**, the computing device can determine, based on input data associated with a user of the computing device, an intent associated with performing an action on the computing device. For example, the user input **205** described in FIG. 2 can be an example of the input data at **502**.

(83) In some implementations, the intent is determined at **502** with the application-level assistant component.

(84) At **504**, the computing device can invoke, using an application-level assistant component configured to provide an application-level interface on the computing device, execution of an application. For example, the application-level assistant component **220** described in FIG. 2 can be an example of the application-level assistant component at **504**. Additionally, the mobile application **240** described in FIG. 2 can be an example of the application in **504**.

(85) At **506**, the computing device can transmit, to a system-level assistant component configured to provide a system-level interface with the computing device, instructions to interface with an actionable component of the application. The instructions can be transmitted using a trusted intermediary component configured to facilitate an interface between the system-level assistant component and the application-level assistant component.

(86) For example, the system-level assistant component **210** described in FIG. 2 can be an example of the system-level assistant component at **506**. An actionable component of the application can include an action component associated with a first page **245**, second page **250**, first activity, second activity, and so on of the mobile application **240**. Additionally, the trusted intermediary component **230** described in FIG. 2 can be an example of the trusted intermediary component at

506.

(87) In some implementations, the trusted intermediary component can include one or more application program interfaces (APIs) between the system-level assistant component **210** and the application-level assistant component **220**.

(88) In some implementations, the actionable component at **506** can be an API.

(89) At **508**, the computing device can engage, using the system-level assistant component, the actionable component of the application to perform the action. For example, the system-level assistant component **210** described in FIG. 2 can be an example of the system-level assistant component at **508**.

(90) In some implementations, the actionable component at **506** can be an input field, and the engagement of the actionable component at **508** can include inputting a value in the input field.

(91) In some implementations, the actionable component at **506** can be a rendered output, and the engagement of the actionable component at **508** can include parsing rendered output to identify input fields.

(92) In some implementations, the system-level assistant component can generate an input having data unavailable to the application-level assistant component. For example, the action performed at **508** can include the system-level assistant component **210** generating an input that includes data (e.g., user data) that is unavailable to the application-level assistant component **220**.

(93) In some implementations, the engagement performed at **508** can include determining missing information and generating a follow-up query. For example, the system-level assistant component **210** can determine that the value to be inputted in the input field is unknown, and therefore the automation system **200** can ask a follow-up query to the user.

(94) In some implementations, when missing information is determined at **508**, method **500** can further include initiating a follow-up query requesting the missing information. For example, method **500** can further include the system-level assistant component generating a follow-up query. Furthermore, the system-level assistant component can parse a response to the follow-up query. Moreover, method **500** can further include the system-level assistant component initially parsing a response to follow-up query and then handing off to the application-level assistant component.

(95) In some implementations, the system-level assistant component can perform an automatic speech recognition (ASR) technique and application-level assistant component can perform a natural-language understanding (NLU) technique.

(96) In some implementations, the application-level assistant component is a third-party component.

(97) In some implementations, method **500** further includes the computing device receiving content item comprising application-level assistant component, the content item obtained from a real-time content selection component of a remote server (e.g., server computing system **130**). For example, a custom chatbot experiences for businesses can be performed by the computing device by receiving content items from a real-time content selection component of the server computing system **130**.

(98) In some implementations, method **500** further includes the computing device retrieving, using application-level assistant component, a content item obtained from a real-time content selection component of a remote server, the content item comprising the application.

(99) In some implementations, method **500** can further include retrieving, using the application-level assistant component, a content item obtained from a real-time content selection component of a remote server, the content item comprising the application.

(100) In some implementations, the system-level assistant component can be running as part of an operating system of the computing device.

(101) In some implementations, the automation system **200** can automate the journey using a hierarchical approach. For example, the application-level assistant component **220** can handle the inter-application journeys (e.g., opening a mobile application, sending an intent, or passing

parameters between mobile applications), while the system-level assistant component **210** can handle the intra-application journeys (e.g., performing one or more UI actions on a specific activity of a mobile application, performing one or more UI actions on a specific page of a mobile application).

(102) FIG. **6** depicts a flow chart diagram of a human-in-the-loop example method for automating an action to be performed on a mobile application according to example embodiments of the present disclosure. Although FIG. **6** depicts steps performed in a particular order for purposes of illustration and discussion, the methods of the present disclosure are not limited to the particularly illustrated order or arrangement. The various steps of method **600** can be omitted, rearranged, combined, and/or adapted in various ways without deviating from the scope of the present disclosure.

(103) Continuing with method **500**, the engaging the actionable component can include the automation system **200** determining missing information. The method **500** can further include the operations of method **600**.

(104) In some implementations, the system-level assistant component **210** or the application-level assistant component **220** can generate a follow-up query (e.g., a question for the user to answer). The follow-up query can be generated in response to a determination at **508** that information to be inputted is missing.

(105) At **602**, the computing device (e.g., automation system **200**) can parse, using the system-level assistant component, a response to the follow-up query.

(106) At **604**, the computing device can determine, by performing an automatic speech recognition (ASR) technique using the system-level assistant component, that the response to the follow-up query is unknown. For example, the system-level assistant component **210** cannot determine the intent of the response or the system-level assistant component **210** cannot determine a parameter from the response.

(107) At **606**, in response to the determination that the response is unknown at **604**, the computing device can transmit, using the trusted intermediary **230**, the response to the application-level assistant component **220**. The response can be securely transmitted, using the trusted intermediary **230**, from the system-level assistant component **210** to the application-level assistant component **220** so that the user does not have to repeat the answer to the query twice.

(108) At **608**, the computing device can analyze, by performing a natural-language understanding (NLU) technique using the application-level assistant component, the response to the follow-up query to determine a parameter.

(109) At **610**, the computing device can transmit, using the trusted intermediary, the parameter to the system-level assistant component.

(110) At **612**, the computing device can engage, using the system-level assistant component, the actionable component of the application by inputting the parameter in an input field.

(111) In some implementations, the operations in method **600** can further include modifying a sequence of tasks being performed by the system-level assistant component based on a response to the follow-up query. For example, the human-in-the-loop implementations of the techniques described herein allows for the modification of tasks (e.g., path for performing an activity can be modified) based on an input from the user.

(112) FIG. **7** depicts a flow chart diagram of another example method for automating an action to be performed on a mobile application according to example embodiments of the present disclosure. Although FIG. **7** depicts steps performed in a particular order for purposes of illustration and discussion, the methods of the present disclosure are not limited to the particularly illustrated order or arrangement. The various steps of method **700** can be omitted, rearranged, combined, and/or adapted in various ways without deviating from the scope of the present disclosure.

(113) In some implementations, a computing device (e.g., user computing device **102**, server computing system **130**, automation system **200**, speaker **305**, client device **340**) can include an

automation system (e.g., automation system **200**). For example, the first system can be the system-level assistant component **210** in FIG. 2. The first system can perform user interface (UI) actions on a plurality of mobile applications. The second system can include a virtual assistant application, such as the application-level assistant component **220** in FIG. 2. The third system can have one or more application program interfaces (APIs) between the first system and the second system, such as the trusted intermediary component **230** in FIG. 2. Additionally, the automation system can include one or more processors, and a memory storing instructions that when executed by the one or more processors cause the computer device to perform operations described in method **700**.

(114) At **702**, the computing device can receive, via an interface of the computing device, a user request (e.g., user input data) having audio data. The user input **205** in FIG. 2 can be an example of the user request received at **702**.

(115) At **704**, the computing device can determine, based on the user request, a first query and a second query. For example, the first query and the second query can be obtained from the plurality of UI scripts **270** in FIG. 2. In some instances, the computing device can determine (e.g., obtain from the plurality of UI scripts **270** in FIG. 2) a first UI script and a second UI script based on the user request.

(116) For example, the automation system **200** can use the user request to derive the parameters for a first query (e.g., first task, first activity, first action), a second query (e.g., second task), a third query (e.g., third task), and any subsequent queries (e.g., subsequent tasks). The automation system **200** can also send the data back to the application-level assistant component **220** (e.g., user-space virtual assistant application) so that the application-level assistant component **220** can alter the following tasks or ask relevant clarification questions to the user

(117) At **706**, the computing device can transmit the first query to the first system. The first query can be associated with the first system performing a first UI action on a first mobile application from the plurality of mobile applications. In some implementations, the first UI action can be a low-level action as described in FIG. 2.

(118) For example, the first UI action performed by the first system can be scrolling and/or clicking a button on a page of the first mobile application.

(119) In some implementations, the first system can include a system-level assistant component (e.g., system-level assistant component **210** in FIG. 2) running as part of an operating system of the computing device. The operating system can be software that acts as an interface between computer hardware components and a user.

(120) In some implementations, the computing device can be a mobile device. Additionally, the first system can be part of the middleware of the mobile device. The middleware is the software that lies between an operating system and the applications running on it.

(121) At **708**, the computing device can transmit, to the second system, the second query and a first parameter generated by the first UI action to the second system. The second query can be determined at **704**. The first parameter and the second query can be transmitted using the third system. For example, the automation system **200** can transmit the first parameter from the system-level assistant component **210** to the application-level assistant component **220** by using the trusted intermediary component **230** between the system-level assistant component **210** and the application-level assistant component **220**.

(122) In some implementations, the trusted intermediary component **230** includes a middleware API. In some implementations, the trusted intermediary component **230** includes an operating system kernel API. In some implementations, the trusted intermediary component **230** includes a direct remote procedure call (RPC)-based API.

(123) At **710**, the computing device can perform, using the second system, a second action (e.g., high-level) based on the second query and the first parameter. For example, the automation system **200** can perform a high-level action by using the application-level assistant component **220**. Examples of high-level actions are described in FIG. 2. Additionally, the application-level assistant

component **220** can perform the high-level action based on the second query (e.g., a second UI script that is obtained from the plurality of UI scripts **270** in FIG. 2).

(124) In some implementations, the first system can have system-level privileges that are higher than privileges associated with the application-level assistant component (e.g., application-level assistant component **220**) application of the second system. In some implementations, the high-level action performed by the second system is to play a video.

(125) In some implementations, the virtual assistant application is stored in virtual memory associated with a user-space. In some implementations, the parameters transferred between the first system and the second system are only transmitted using the third system.

(126) In some implementations, method **700** can also include the computing device obtaining user consent before the first system performs the first UI action on the first mobile application. Additionally, the computing device can display, while the first system is performing the first UI action, an indicator on a display of the computing device that the first system is controlling the computing device.

(127) FIG. **8** depicts a flow chart diagram of an example method of a human-in-the-loop implementation for modifying a query, according to example embodiments of the present disclosure. Although FIG. **8** depicts steps performed in a particular order for purposes of illustration and discussion, the methods of the present disclosure are not limited to the particularly illustrated order or arrangement. The various steps of method **800** can be omitted, rearranged, combined, and/or adapted in various ways without deviating from the scope of the present disclosure.

(128) At **802**, continuing with method **700**, the computing device can perform, using the second system, a high-level action based on the second query and the first parameter. The operation **602** can be similar to operation **710** of method **700**. The high-level action can be to ask a clarifying question to the user associated with the user input received at **702**.

(129) At **804**, the computing device can determine, based on the user request, a third query.

(130) At **806**, the computing device can receive, via the interface of the computing device, an answer in response to the question.

(131) At **808**, the computing device can modify the third query based on the answer.

(132) In some implementations, the modification of the third query includes altering one or more parameters being transmitted between the first system and the second system.

(133) In some implementations, the modification of the third query includes altering a sequence path associated with subsequent queries to be performed by the first system or the second system.

(134) FIG. **9** depicts a flow chart diagram of an example method to securely transfer data between two mobile applications, according to example embodiments of the present disclosure. Although FIG. **9** depicts steps performed in a particular order for purposes of illustration and discussion, the methods of the present disclosure are not limited to the particularly illustrated order or arrangement. The various steps of method **900** can be omitted, rearranged, combined, and/or adapted in various ways without deviating from the scope of the present disclosure.

(135) At **902**, continuing with method **700**, the computing device can perform, using the second system, a high-level action based on the second query and the first parameter. The operation **702** can be similar to operation **710** of method **700**. The high-level action can for the second system to open a second mobile application from the plurality of mobile applications.

(136) At **904**, the computing device can determine, based on the user request, a third query.

(137) At **906**, the computing device can transmit, using the third system, a second parameter generated by the second system to the first system.

(138) At **908**, the computing device can transmit the third query to the first system. The third query can be associated with the first system performing a second UI action on the second mobile application.

Additional Disclosure

(139) The technology discussed herein makes reference to servers, databases, software applications, and other computer-based systems, as well as actions taken, and information sent to and from such systems. The inherent flexibility of computer-based systems allows for a great variety of possible configurations, combinations, and divisions of tasks and functionality between and among components. For instance, processes discussed herein can be implemented using a single device or component or multiple devices or components working in combination. Databases and applications can be implemented on a single system or distributed across multiple systems. Distributed components can operate sequentially or in parallel.

(140) While the present subject matter has been described in detail with respect to various specific example embodiments thereof, each example is provided by way of explanation, not limitation of the disclosure. Those skilled in the art, upon attaining an understanding of the foregoing, can readily produce alterations to, variations of, and equivalents to such embodiments. Accordingly, the subject disclosure does not preclude inclusion of such modifications, variations and/or additions to the present subject matter as would be readily apparent to one of ordinary skill in the art. For instance, features illustrated or described as part of one embodiment can be used with another embodiment to yield a still further embodiment. Thus, it is intended that the present disclosure covers such alterations, variations, and equivalents.

(141) The depicted and/or described steps are merely illustrative and can be omitted, combined, and/or performed in an order other than that depicted and/or described; the numbering of depicted steps is merely for ease of reference and does not imply any particular ordering is necessary or preferred.

(142) The functions and/or steps described herein can be embodied in computer-usable data and/or computer-executable instructions, executed by one or more computers and/or other devices to perform one or more functions described herein. Generally, such data and/or instructions include routines, programs, objects, components, data structures, or the like that perform particular tasks and/or implement particular data types when executed by one or more processors in a computer and/or other data-processing device. The computer-executable instructions can be stored on a computer-readable medium such as a hard disk, optical disk, removable storage media, solid-state memory, read-only memory (ROM), random-access memory (RAM), or the like. As will be appreciated, the functionality of such instructions can be combined and/or distributed as desired. In addition, the functionality can be embodied in whole or in part in firmware and/or hardware equivalents, such as integrated circuits, application-specific integrated circuits (ASICs), field-programmable gate arrays (FPGAs), or the like. Particular data structures can be used to implement one or more aspects of the disclosure more effectively, and such data structures are contemplated to be within the scope of computer-executable instructions and/or computer-usable data described herein.

(143) Although not required, one of ordinary skill in the art will appreciate that various aspects described herein can be embodied as a method, system, apparatus, and/or one or more computer-readable media storing computer-executable instructions. Accordingly, aspects can take the form of an entirely hardware embodiment, an entirely software embodiment, an entirely firmware embodiment, and/or an embodiment combining software, hardware, and/or firmware aspects in any combination.

(144) As described herein, the various methods and acts can be operative across one or more computing devices and/or networks. The functionality can be distributed in any manner or can be located in a single computing device (e.g., server, client computer, user device, or the like).

(145) Aspects of the disclosure have been described in terms of illustrative embodiments thereof. Numerous other embodiments, modifications, and/or variations within the scope and spirit of the appended claims can occur to persons of ordinary skill in the art from a review of this disclosure. For example, one of ordinary skill in the art can appreciate that the steps depicted and/or described can be performed in other than the recited order and/or that one or more illustrated steps can be

optional and/or combined. Any and all features in the following claims can be combined and/or rearranged in any way possible.

(146) While the present subject matter has been described in detail with respect to various specific example embodiments thereof, each example is provided by way of explanation, not limitation of the disclosure. Those skilled in the art, upon attaining an understanding of the foregoing, can readily produce alterations to, variations of, and/or equivalents to such embodiments. Accordingly, the subject disclosure does not preclude inclusion of such modifications, variations, and/or additions to the present subject matter as would be readily apparent to one of ordinary skill in the art. For instance, features illustrated and/or described as part of one embodiment can be used with another embodiment to yield a still further embodiment. Thus, it is intended that the present disclosure covers such alterations, variations, and/or equivalents.

(147) The following is a list of aspects of the present disclosure which may or may not be claimed:

(148) Aspect 1. A computing device, comprising: one or more processors; and one or more memory devices storing instructions that are executable by the one or more processors to cause the one or more processors to perform operations, the operations comprising: determining, based on input data associated with a user of the computing device, an intent associated with performing an action on the computing device; invoking, using an application-level assistant component configured to provide an application-level interface on the computing device, execution of an application; transmitting, to a system-level assistant component configured to provide a system-level interface with the computing device, instructions to interface with an actionable component of the application, wherein the instructions are transmitted using a trusted intermediary component configured to facilitate an interface between the system-level assistant component and the application-level assistant component; and engaging, using the system-level assistant component, the actionable component of the application to perform the action.

(149) Aspect 2. The computing device of aspect 1, wherein the actionable component comprises an input field, and wherein engagement of the actionable component comprises inputting a value in the input field.

(150) Aspect 3. The computing device of aspect 1 or aspect 2, wherein the actionable component comprises a rendered output, and wherein engagement of the actionable component comprises parsing the rendered output to identify input fields.

(151) Aspect 4. The computing device of any preceding aspect, wherein the intent is determined using the application-level assistant component.

(152) Aspect 5. The computing device of any preceding aspect, wherein the trusted intermediary component comprises an application program interface (API).

(153) Aspect 6. The computing device of any preceding aspect, wherein the system-level assistant component generates an input comprising data unavailable to the application-level assistant component.

(154) Aspect 7. The computing device of any preceding aspect, wherein engagement of the actionable component comprises determining missing information, and the operations further comprising: initiating a follow-up query requesting the missing information.

(155) Aspect 8. The computing device of aspect 7, wherein the system-level assistant component generates the follow-up query.

(156) Aspect 9. The computing device of aspect 7 or aspect 8, the operations further comprising: parsing, using the system-level assistant component, a response to the follow-up query.

(157) Aspect 10. The computing device of aspect 9, the operations further comprising: determining, by performing an automatic speech recognition (ASR) technique using the system-level assistant component, that the response to the follow-up query is unknown; transmitting, using the trusted intermediary, the response to the application-level assistant component; and analyzing, by performing a natural-language understanding (NLU) technique using the application-level assistant component, the response to the follow-up query to determine a parameter.

(158) Aspect 11. The computing device of aspect 10, the operations further comprising: transmitting, using the trusted intermediary, the parameter to the system-level assistant component; and engaging, using the system-level assistant component, the actionable component of the application by inputting the parameter in an input field.

(159) Aspect 12. The computing device of any of aspects 7 to 11, the operations further comprising: modifying a sequence of tasks being performed by the system-level assistant component based on a response to the follow-up query.

(160) Aspect 13. The computing device of any preceding aspect, wherein the system-level assistant component performs an automatic speech recognition (ASR) technique and application-level assistant component performs a natural-language understanding (NLU) technique.

(161) Aspect 14. The computing device of any preceding aspect, wherein the application-level assistant component comprises a third-party assistant component.

(162) Aspect 15. The computing device of any preceding aspect, the operations further comprising: retrieving, using the application-level assistant component, a content item obtained from a real-time content selection component of a remote server, the content item comprising the application.

(163) Aspect 16. The computing device of any preceding aspect, wherein the system-level assistant component is running as part of an operating system of the computing device or a middleware of the computing device.

(164) Aspect 17. The computing device of any preceding aspect, wherein the system-level assistant component has system-level privileges, wherein the system-level privileges includes one or more privileges not available to the application-level assistant component.

(165) Aspect 18. The computing device of any preceding aspect, wherein the application-level assistant component is opening the application, and wherein the system-level assistant component is performing a user-interface action associated with an activity of the application.

(166) Aspect 19. The computing device of any preceding claim, wherein the computing device is a user computing device, and wherein the application-level assistant component and the system level assistant component are implemented by the user computing device.

(167) Aspect 20. A computer-implemented method comprising: determining, based on input data associated with a user of a computing device, an intent associated with performing an action on the computing device; invoking, using an application-level assistant component configured to provide an application-level interface on the computing device, execution of an application; transmitting, to a system-level assistant component configured to provide a system-level interface with the computing device, instructions to interface with an actionable component of the application, wherein the instructions are transmitted using a trusted intermediary component configured to facilitate an interface between the system-level assistant component and the application-level assistant component; and engaging, using the system-level assistant component, the actionable component of the application to perform the action.

(168) Aspect 21. The computer-implemented method of aspect 20, wherein the actionable component comprises an input field, and wherein engagement of the actionable component comprises inputting a value in the input field.

(169) Aspect 22. The computer-implemented method of aspect 20 or aspect 21, wherein the actionable component comprises a rendered output, and wherein engagement of the actionable component comprises parsing the rendered output to identify input fields.

(170) Aspect 23. The computer-implemented method of any of aspects 20 to 22, wherein the intent is determined using the application-level assistant component.

(171) Aspect 24. The computer-implemented method of any of aspects 20 to 23, wherein the trusted intermediary component comprises an application program interface (API).

(172) Aspect 25. The computer-implemented method of any of aspects 20 to 24, wherein the system-level assistant component generates an input comprising data unavailable to the application-level assistant component.

(173) Aspect 26. The computer-implemented method of any of aspects 20 to 25, wherein engagement of the actionable component comprises determining missing information, and the operations further comprising: initiating a follow-up query requesting the missing information.

(174) Aspect 27. The computer-implemented method of aspect 26, wherein the system-level assistant component generates the follow-up query.

(175) Aspect 28. The computer-implemented method of aspect 26 or aspect 27, the operations further comprising: parsing, using the system-level assistant component, a response to the follow-up query.

(176) Aspect 29. The computer-implemented method of aspect 28, the operations further comprising: determining, by performing an automatic speech recognition (ASR) technique using the system-level assistant component, that the response to the follow-up query is unknown; transmitting, using the trusted intermediary, the response to the application-level assistant component; and analyzing, by performing a natural-language understanding (NLU) technique using the application-level assistant component, the response to the follow-up query to determine a parameter.

(177) Aspect 30. The computer-implemented method of aspect 29, the operations further comprising: transmitting, using the trusted intermediary, the parameter to the system-level assistant component; and engaging, using the system-level assistant component, the actionable component of the application by inputting the parameter in an input field.

(178) Aspect 31. The computer-implemented method of any of aspects 27 to 30, the operations further comprising: modifying a sequence of tasks being performed by the system-level assistant component based on a response to the follow-up query.

(179) Aspect 32. The computer-implemented method of any of aspects 20 to 31, wherein the system-level assistant component performs an automatic speech recognition (ASR) technique and application-level assistant component performs a natural-language understanding (NLU) technique.

(180) Aspect 33. The computer-implemented method of any of aspects 20 to 32, wherein the application-level assistant component comprises a third-party assistant component.

(181) Aspect 34. The computer-implemented method of any of aspects 20 to 33, the operations further comprising: retrieving, using the application-level assistant component, a content item obtained from a real-time content selection component of a remote server, the content item comprising the application.

(182) Aspect 35. The computer-implemented method of any of aspects 20 to 34, wherein the system-level assistant component is running as part of an operating system of the computing device or a middleware of the computing device.

(183) Aspect 36. The computer-implemented method of any of aspects 20 to 35, wherein the system-level assistant component has system-level privileges, wherein the system-level privileges includes one or more privileges not available to the application-level assistant component.

(184) Aspect 37. The computer-implemented method of any of aspects 20 to 36, wherein the application-level assistant component is opening the application, and wherein the system-level assistant component is performing a user-interface action associated with an activity of the application.

(185) Aspect 38. The computer-implemented method of any of aspects 20 to 37, wherein the computing device is a user computing device, and wherein the application-level assistant component and the system level assistant component are implemented by the user computing device.

(186) Aspect 39. One or more non-transitory computer-readable media comprising instructions that when executed by one or more computing devices cause the one or more computing devices to perform operations comprising: determining, based on input data associated with a user of the computing device, an intent associated with performing an action on the computing device; invoking, using an application-level assistant component configured to provide an application-level

interface on the computing device, execution of an application; transmitting, to a system-level assistant component configured to provide a system-level interface with the computing device, instructions to interface with an actionable component of the application, wherein the instructions are transmitted using a trusted intermediary component configured to facilitate an interface between the system-level assistant component and the application-level assistant component; and engaging, using the system-level assistant component, the actionable component of the application to perform the action.

(187) Aspect 40. One or more computer readable media comprising instructions that when executed by one or more computing devices cause the one or more computing devices to perform the method according to any of aspects 20 to 38.

Claims

1. A computing device, comprising: one or more processors; and one or more memory devices storing instructions that are executable by the one or more processors to cause the one or more processors to perform operations, the operations comprising: determining, based on input data associated with a user of the computing device, an intent associated with performing an action on the computing device; invoking, using an application-level assistant component configured to provide an application-level interface on the computing device, execution of an application; transmitting, to a system-level assistant component configured to provide a system-level interface with the computing device, instructions to interface with an actionable component of the application, wherein the instructions are transmitted using a trusted intermediary component configured to facilitate an interface between the system-level assistant component and the application-level assistant component, wherein the trusted intermediary component comprises an application program interface (API); and engaging, using the system-level assistant component, the actionable component of the application to perform the action.
2. The computing device of claim 1, wherein the API is a middleware API, an operating system kernel API, or a direct remote procedure call (RPC)-based API.
3. The computing device of claim 1, wherein the actionable component comprises an input field, and wherein engagement of the actionable component comprises inputting a value in the input field.
4. The computing device of claim 1, wherein the actionable component comprises a rendered output, and wherein engagement of the actionable component comprises parsing the rendered output to identify input fields.
5. The computing device of claim 1, wherein the intent is determined using the application-level assistant component.
6. The computing device of claim 1, wherein the system-level assistant component generates an input comprising data unavailable to the application-level assistant component.
7. The computing device of claim 1, wherein engagement of the actionable component comprises determining missing information, and the operations further comprising: initiating a follow-up query requesting the missing information.
8. The computing device of claim 7, wherein the system-level assistant component generates the follow-up query.
9. The computing device of claim 7, the operations further comprising: parsing, using the system-level assistant component, a response to the follow-up query.
10. The computing device of claim 9, the operations further comprising: determining, by performing an automatic speech recognition (ASR) technique using the system-level assistant component, that the response to the follow-up query is unknown; transmitting, using the trusted intermediary, the response to the application-level assistant component; and analyzing, by performing a natural-language understanding (NLU) technique using the application-level assistant

component, the response to the follow-up query to determine a parameter.

11. The computing device of claim 10, the operations further comprising: transmitting, using the trusted intermediary, the parameter to the system-level assistant component; and engaging, using the system-level assistant component, the actionable component of the application by inputting the parameter in an input field.

12. The computing device of claim 7, the operations further comprising: modifying a sequence of tasks being performed by the system-level assistant component based on a response to the follow-up query.

13. The computing device of claim 1, wherein the system-level assistant component performs an automatic speech recognition (ASR) technique and application-level assistant component performs a natural-language understanding (NLU) technique.

14. The computing device of claim 1, wherein the application-level assistant component comprises a third-party assistant component.

15. The computing device of claim 1, the operations further comprising: retrieving, using the application-level assistant component, a content item obtained from a real-time content selection component of a remote server, the content item comprising the application.

16. The computing device of claim 1, wherein the system-level assistant component is running as part of an operating system of the computing device or a middleware of the computing device.

17. The computing device of claim 1, wherein the system-level assistant component has system-level privileges, wherein the system-level privileges includes one or more privileges not available to the application-level assistant component.

18. The computing device of claim 1, wherein the application-level assistant component is opening the application, and wherein the system-level assistant component is performing a user-interface action associated with an activity of the application.

19. The computing device of claim 1, wherein the computing device is a user computing device, and wherein the application-level assistant component and the system level assistant component are implemented by the user computing device.

20. A computer-implemented method comprising: determining, based on input data associated with a user of a computing device, an intent associated with performing an action on the computing device; invoking, using an application-level assistant component configured to provide an application-level interface on the computing device, execution of an application; transmitting, to a system-level assistant component configured to provide a system-level interface with the computing device, instructions to interface with an actionable component of the application, wherein the instructions are transmitted using a trusted intermediary component configured to facilitate an interface between the system-level assistant component and the application-level assistant component, wherein the trusted intermediary component comprises an application program interface (API); and engaging, using the system-level assistant component, the actionable component of the application to perform the action.

21. One or more non-transitory computer-readable media comprising instructions that when executed by one or more computing devices cause the one or more computing devices to perform operations comprising: determining, based on input data associated with a user of the computing device, an intent associated with performing an action on the computing device; invoking, using an application-level assistant component configured to provide an application-level interface on the computing device, execution of an application; transmitting, to a system-level assistant component configured to provide a system-level interface with the computing device, instructions to interface with an actionable component of the application, wherein the instructions are transmitted using a trusted intermediary component configured to facilitate an interface between the system-level assistant component and the application-level assistant component, wherein the trusted intermediary component comprises an application program interface (API); and engaging, using the

system-level assistant component, the actionable component of the application to perform the action.
